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Quantum Repeater Chains via Cavity–Magnon for Scalable Quantum Networks

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ABSTRACT Scalable quantum networks require quantum repeaters to overcome major challenges, such as photon loss and decoherence, in long-distance quantum communication. In this article, we present a cavity–magnon quantum repeater architecture that exploits the frequency tunability and hybrid integration capabilities of magnonic platforms to enable efficient entanglement swapping across multihop networks. The coherence budget is explicitly incorporated via the dissipation and dephasing terms in our noise model, which sets the effective entanglement-storage window available for multihop swapping. Through comprehensive numerical simulations with realistic experimental parameters, we analyze system performance across diverse deployment scenarios and network scales, examining both short-range and long-distance implementations. We identify critical factors influencing performance and scalability, demonstrating that cavity–magnon systems represent a viable and promising quantum repeater platform with significant integration advantages over existing quantum memory technologies.

INDEX TERMS Quantum communication, distributed quantum computing (DQC), quantum networks, quantum key distribution (QKD).

I. INTRODUCTION

Recent progress in quantum communication promises immense potential to transform the way we secure and transfer information by harnessing uniquely quantum resources, such as entanglement and superposition. The long-term vision of quantum networks and a global quantum internet is to connect distant quantum processors and local quantum networks to enable applications in cryptography, distributed computing, and quantum-enhanced sensing [1]. However, realizing this vision faces fundamental technical challenges, most notably photon loss and decoherence over realistic channels.

Building quantum repeaters is not an easy feat, as they require quantum memories, which are challenging to build in their own right [2]. In addition, photon loss and decoherence during transmission through optical fibers or aerial links strongly limit long-distance distribution of entanglement and constrain the number and efficiency of local gate operations that can be applied before the stored state degrades. In classical communication networks, losses can be compensated by amplification, but this is not possible in quantum networks due to the no-cloning theorem and the Heisenberg uncertainty principle [3]. Quantum repeaters provide an alternative by dividing the channel into shorter segments, generating entanglement locally, and then extending it via entanglement swapping [4], [5], [6]. In this way, quantum information can be transferred end-to-end without requiring a single photon to traverse the full distance, mitigating channel loss while maintaining coherence.

Considerable progress has already been made in generating entangled correlations across a wide range of platforms, including cavity quantum electrodynamics (Cavity-QED) [7], [8], [9], circuit quantum electrodynamics (Circuit-QED) [10], [11], nuclear magnetic resonance [12], spintronics [13], and photonic systems [14]. Many of these platforms have also been explored as building blocks for quantum repeaters [15], [16], each with distinct control requirements and hardware constraints. Among these platforms, cavity magnonics is a promising candidate due to its tunable magnon–photon coupling, low intrinsic damping, and compatibility with superconducting microwave circuits,

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nitrogen-vacancy (NV) centers in diamond, and optical transduction schemes, making it well suited for hybrid quantum architectures [17], [18], [19]. In representative low-temperature Yttrium–Iron–Garnet (YIG)-based cavity–magnonic devices, the magnon linewidth is typically on the order of 1 MHz, corresponding to coherence times in the range of hundreds of nanoseconds to submicroseconds [20]. In our simulations, the adopted noise parameters correspond to a cavity decay time of 159 ns and a cavity coherence time of 199 ns, while the magnon mode has a decay time of 318 ns and a coherence time of 289 ns.

Considerable experimental milestones for quantum networks have been achieved over the past few years, with quantum repeater technology rapidly advancing from laboratory prototypes to deployable infrastructure. In 2020, the USTC group in China achieved a significant milestone by extending memory-to-memory entanglement over 12.5 km, achieving city-scale distribution using atomic-ensemble memories [21]. In 2021, a team at Delft linked three NV-center nodes through entanglement swapping over a separation of 1.3 km [22]. Building on this, in 2022, the same group succeeded in teleporting quantum information across nonneighboring nodes [23], marking a major step forward. In 2023, researchers at the University of Innsbruck demonstrated entanglement swapping over a fiber channel using an intertelecom wavelength, generating heralded entanglement between two distant parties over 50 km [15]. Furthermore, recent demonstrations on existing metropolitan networks, such as the AWS-Harvard link in Boston (about 35 km), the Argonne 84 km Chicago test loop, and the Qunnect 16 km GothamQ network in New York, proved the robustness of quantum hardware outside the laboratory. In addition, various national initiatives under the European Union and elsewhere have established testbeds for quantum communication [24], [25]. Collectively, these achievements have charted a firm trajectory toward repeater-enabled quantum networks.

Recently, cavity magnonics has emerged as a hybrid architecture enabling integration of Cavity-QED or Circuit-QED with YIG-magnon systems [17], and the integration of superconducting circuits with YIG magnons has also been explored [26], [27]. More broadly, hybrid quantum systems based on magnonics have been proposed and demonstrated as a platform for interfacing magnons with other physical degrees of freedom in integrated architectures [28]. This motivates engineered hardware stacks that combine high-fidelity microwave processing with magnon-enabled storage and interfacing. Superconducting qubits are excellent local processors, but a repeater node also benefits from a separate, frequency-tunable interface/memory mode that can be efficiently loaded and retrieved and that couples strongly to cavities and transduction hardware. Magnons provide such a bosonic interface via strong, tunable cavity coupling that enables SWAP-based state transfer and supports a modular stack that separates processing (qubits) from buffering and interfacing for multihop networking [17], [20], [29]. Magnons, by contrast, provide tunable coupling, access to multiple coupling regimes, and can serve as short-term interface memories and interaction mediators in hybrid superconducting-magnonic nodes, enabling coherent transfer between processing and transduction stages [26], [27]. Furthermore, YIG-based magnon modes can exhibit low intrinsic damping and strong, tunable coupling to microwave cavities, which makes them attractive as short-term interface memories and interaction mediators in hybrid quantum architectures [20].

The ability to establish remote entanglement between distant quantum objects, magnons in our case, is a fundamental step toward large-scale quantum networks. Such networks remain constrained by finite entanglement generation rates and limited coherence budgets [30], [31]. Accordingly, our model explicitly incorporates both relaxation and dephasing processes, which bound the allowable buffering time for heralded entanglement distribution and motivate the use of multiplexing and low-latency feedforward to reduce waiting time. This approach is uniquely advantageous in quantum networks, where communication resources can be prepared in advance.

A. CONTRIBUTIONS

In this article, we focus on using a hybrid cavity magnonics structure to generate entanglement over long distances, thereby enabling flexible quantum communication. Our cavity–magnonic system works as a quantum repeater in which remote entanglement is generated and extended via successive entanglement swapping operations. We use magnonic nodes because they are flux-tunable, couple strongly to cavities, and can interface with both optical and microwave systems [32], [33]. In our simulations, we model coherence using dissipation and dephasing rates, which determine the effective memory times used in the performance evaluation. Our proposed setup consists of identical elementary links that connect these individual setups remotely. In our study, we investigate two scenarios related to communication infrastructure. The first scenario involves a compact setup that can be integrated onto a chip, allowing for implementation within the microwave domain. The second scenario addresses larger systems, such as those deployed at a city scale, where microwave-to-optical converters are necessary to bridge the connection with optical communication networks.

We benchmark the proposed cavity-magnon repeater using a composite figure of merit that couples entanglement quality, as captured by concurrence and conditional Bell-state fidelity, with the distribution rate embodied in the heralded success probability, by simulating these metrics under realistic dissipation, fiber attenuation and multiplexing in both chip-scale and metro-scale scenarios. All numerical results use parameter sets drawn directly from state-of-the-art microwave-cavity, optical-conversion and dense-multiplexing experiments, ensuring that our simulations reflect realistic hardware performance while isolating the architecture-level design tradeoffs. This benchmarking

also helps us pinpoint the main limits in each scenario, including conversion loss and fiber loss in the metro-scale case.

B. ORGANIZATION

The rest of this article is organized as follows. Section II presents the preliminaries required to quantify the generated states and to describe how cavity-magnon entanglement is prepared, including the figures of merit (concurrence and fidelity) and the cavity-magnon interaction and noise model. Section III describes the elementary gate operations used in the protocol under realistic noise. Section IV describes the entanglement-swapping protocol and shows how repeated swapping operations form a cavity-magnonic repeater chain capable of distributing heralded Bell pairs over long distances. Section V presents the numerical results for chip-scale and metro-scale scenarios and discusses the key performance trends and limitations. Finally, Section VI concludes this article.

II. PRELIMINARIES

In this section, we review the fundamentals of magnon–photon interactions in open quantum systems, establishing the theoretical foundation for cavity–magnon entanglement generation. We begin by introducing the quantum resource metrics, namely concurrence and fidelity, which serve as benchmarks for evaluating our proposed quantum repeater scheme. We then examine cavity–magnon entanglement generation, beginning with the ideal unitary dynamics and progressing to realistic open-system evolution under environmental noise, with numerical simulations demonstrating high-fidelity Bell state preparation using experimentally relevant parameters.

A. QUANTUM RESOURCES

In quantum communication networks, the quantification of nonclassical correlations is vital to assess protocol performance and ensure reliable entanglement distribution over the noisy channel. Here, the resource theories provide a formal framework for evaluating these quantum features, offering resource monotones that measure the utility of quantum states under a given set of allowed operations. In this work, in order to evaluate the performance of our proposed entanglement swapping-based quantum repeater protocol in the cavity magnonic architecture, we focus on two key resource metrics, concurrence and fidelity.

a) Concurrence: Concurrence is a widely used monotone to quantify quantum entanglement [34], where higher values of concurrence indicate robust quantum entanglement, commonly serving as a critical quantum resource in entanglement-based quantum communication and quantum repeater setups. For a two-qubit density matrix ρ, the concurrence $\mathcal{C}(\rho)$ is defined as

$$\mathcal{C}(\rho) = \max\{0, \lambda_1 - \lambda_2 - \lambda_3 - \lambda_4\} \tag{1}$$

where λ_i represents the square roots of the eigenvalues of the matrix $\rho(\sigma_y \otimes \sigma_y)\rho^*(\sigma_y \otimes \sigma_y)$, arranged in decreasing order, and ρ^* denotes the complex conjugate of the density matrix. The boundaries of the concurrence intervals represent the separable state $\mathcal{C}(t) = 0$ and the maximally entangled state $\mathcal{C}(t) = 1$. In our analysis, we track concurrence dynamically to investigate how entanglement between magnonic subsystems evolves under various interactions for end-to-end entanglement generation under the losses described in the evolution of interaction and swap operations.

b) Fidelity: Fidelity is a fundamental metric in quantum information and is used to quantify the similarity between two quantum states. For two density matrices ρ and σ, representing the actual and ideal quantum states, respectively, the fidelity of the states is defined as

$$\mathcal{F}(\rho, \sigma) = \left(\mathrm{Tr}\left[\sqrt{\sqrt{\rho}\,\sigma\,\sqrt{\rho}}\right]\right)^2. \tag{2}$$

The fidelity is bounded by $0 \leq \mathcal{F}(\rho, \sigma) \leq 1$, where $\mathcal{F} = 1$ implies that the two density matrices are identical. Fidelity is crucial in assessing the success of quantum operations, such as teleportation, entanglement swapping, and error correction. In practical applications, fidelity helps identify decoherence and noise, serving as a guiding metric for optimizing experimental parameters. In our repeater framework, fidelity is used to evaluate the degree to which the resulting entangled magnonic state approximates the desired Bell state after environmental noise and operational imperfections accumulate at each repeater node.

B. CAVITY–MAGNON ENTANGLEMENT GENERATION

In cavity magnonic systems, the composite system consists of ferrimagnetic materials, such as YIG spheres, in high-quality microwave cavities, which enable the formation of hybridized magnon–photon states through tunable magnetic biasing and offer several advantageous properties for quantum information processing, such as frequency tunability over a wide range, and strong magnon–photon coupling that enables efficient quantum state transduction [17], [35], [36]. In our setup, we consider a simplified model with a single cavity mode and a two-level truncated magnon, for which the total Hamiltonian is given as

$$\mathcal{H}_o = \omega_c c^\dagger c + \omega_m m^\dagger m + g_{mc}\left(m + m^\dagger\right)\left(c + c^\dagger\right). \tag{3}$$

Here, the first two terms are the bare energies of the cavity and magnon modes with $c^\dagger$ (c), $m^\dagger$ (m) their respective creation (annihilation) operators for the cavity and magnon modes, and the last term describes the coherent exchange between the subsystems. The coupling strength is $g_{mc} = \gamma\sqrt{\hbar\omega_c\mu_0 S/2V_c}$ where γ is the gyromagnetic ratio, μ_0 is the vacuum permeability, V_c is the cavity mode volume, and S is the total spin of the magnon ensemble. Assuming perfect spatial overlap between the cavity magnetic field and the magnon mode, and working near resonance ($\omega_c \approx \omega_m$), the rotating-wave approximation yields the effective Hamiltonian in the regime $g_{mc} \ll \omega_c, \omega_m$ as

$$\mathcal{H}_i = g_{mc}(m^\dagger c + c^\dagger m). \tag{4}$$

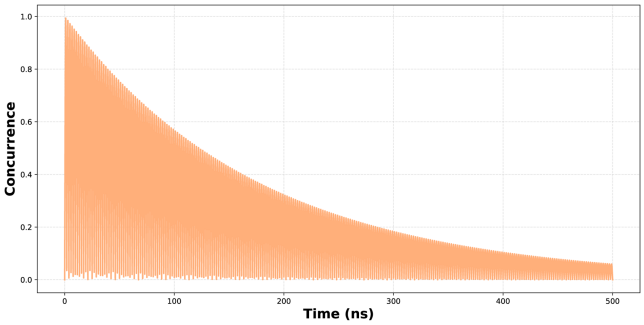


FIGURE 1. Concurrence as a function of time for the cavity-magnon dynamics under Lindblad evolution. Simulation parameters are listed in Table 1.

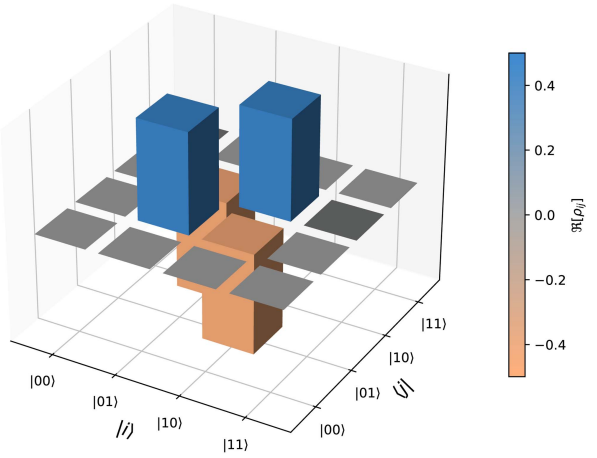


FIGURE 2. Final two-qubit density matrix shown as a 3-D histogram, obtained from Lindblad evolution with the parameters listed in Table 1.

For our system, we consider the initial state $|\psi(0)\rangle = |0\rangle_m|1\rangle_c$, where m and c denote the magnon and cavity modes, respectively. For the ideal case, the system's evolution under $\mathcal{H}_i$ is given by $|\psi(t)\rangle = \mathcal{U}(t)|\psi(0)\rangle$, where the time-evolution operator is $\mathcal{U}(t) = \exp(-i\mathcal{H}_i t)$. The evolved state at the interaction time $t = \pi/(4g_{mc})$ is then given by

$$|\psi(t)\rangle = \frac{|0_m, 1_c\rangle - i|1_m, 0_c\rangle}{\sqrt{2}}. \tag{5}$$

Here, $|\psi(t)\rangle$ is a maximally entangled state locally equivalent to the singlet $|\psi^-\rangle$, which can be represented in the density matrix formalism as $\rho = |\psi(t)\rangle\langle\psi(t)|$. For the nonideal case corresponding to realistic scenarios, environmental interactions introduce noise effects to the quantum system.

The Lindblad formalism provides a framework for describing the dynamical evolution of such open quantum systems, incorporating dissipation, decoherence, and other nonunitary effects added phenomenologically. Mathematically, this is expressed as

$$\frac{d\rho}{dt} = \frac{-i}{\hbar}\left[\hat{\mathcal{H}}, \hat{\rho}\right] + \mathcal{D}(\hat{\rho}). \tag{6}$$

Here, the first term represents the unitary evolution of the system described by the density operator $\hat{\rho}$ under the Hamiltonian $\hat{\mathcal{H}}$. The second term describes the dissipative dynamics, which account for energy exchange with the environment [37]. The superoperator $\mathcal{D}(\hat{\rho})$ is given by

$$\mathcal{D}(\hat{\rho}) = \sum_k \left(\mathcal{L}_k \hat{\rho} \mathcal{L}_k^\dagger - \frac{1}{2}\left\{\mathcal{L}_k^\dagger \mathcal{L}_k, \hat{\rho}\right\}\right). \tag{7}$$

Here, the collapse operators $\mathcal{L}_k$ characterize the dephasing and dissipation effects within the cavity and magnon modes, which are expressed as $\mathcal{L}_c = \sqrt{\kappa_d}\hat{c}$, $\mathcal{L}_m = \sqrt{\gamma_d}\,\hat{m}$, for decay rates and $\mathcal{L}_{c_\phi} = \sqrt{\kappa_\phi}\,\hat{c}^\dagger\hat{c}$ and $\mathcal{L}_{m_\phi} = \sqrt{\gamma_\phi}\,\hat{m}^\dagger\hat{m}$, for the dephasing noise effects of the cavity and magnon modes, respectively [37]. To verify the presence of entanglement and quantify the underlying correlations, we evaluate the system's state over the effective interaction period shown in Fig. 1. We obtain the density matrix from the Lindblad master equation solution computed in quantum toolbox in Python (QuTiP). The resulting density matrix, shown in Fig. 2, demonstrates the formation of the Bell type state in (5), and provides a comprehensive visualization of the system's coherence and population under nonideal conditions.

In Fig. 2, the density matrix representation gives the chromatic intensity gradation to quantitatively depict the magnitude of the constituent elements after Bell-state measurement (BSM). Here, we focus specifically on the strong coupling regime, defined by the condition $g_{mc} \gg (\kappa + \gamma)/2$ where $\kappa = \kappa_d + \kappa_\phi$ and $\gamma = \gamma_d + \gamma_\phi$ are the total dissipation of the respective subsystems. Furthermore, we take the resonant frequency of the system to be at $\omega_i = 2\pi \times 10$ GHz, $i \in \{c, m\}$ for both cavity and magnon modes, respectively, with a coupling rate of $g_{mc} = 2\pi \times 130$ MHz, giving the maximum concurrence at approximately 0.97. We assume the microwave cavities at low temperatures with quality factor in the range $(10^4 - 10^6)$ and cavity decay rates of $\kappa_d = 2\pi \times 1$ MHz. The magnon damping rates, primarily affected by intrinsic Gilbert damping in YIG and material imperfections, are taken to be $\gamma_d = 2\pi \times 0.5$ MHz, while pure dephasing rates for both cavity and magnon modes were maintained at $\kappa_\phi = \gamma_\phi = 2\pi \times 0.3$ MHz. Analysis reveals prominent off-diagonal components (magnitude $\approx \pm 0.47$) that indicate substantial quantum coherence and nonclassical correlations between the bipartite subsystems. This coherence manifests with notable phase symmetry, consistent with theoretical predictions of the Bell type state with minimal decoherence effects. The observed matrix structure demonstrates high-fidelity entanglement preparation.

In our setup, we selected an experimentally relevant time window t using achievable parameters as shown in Table 1, thus demonstrating the system's practical potential and robustness. In the subsequent section, we simulate open system dynamics using the Lindblad master equation and obtain the postswap density matrix from the QuTiP solver output. We then analyze this density matrix to benchmark our quantum repeater architecture.

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TABLE 1. Simulation Parameters Used for Cavity–Magnon Dynamics and Concurrence Calculations

Parameter	Symbol	Value
Cavity frequency	$\omega_c/2\pi$	10 GHz
Magnon frequency	$\omega_m/2\pi$	10 GHz
Magnon–cavity coupling	$g_{mc}/2\pi$	130 MHz
Cavity dissipation rate	$\kappa_d/2\pi$	1 MHz
Magnon dissipation rate	$\gamma_d/2\pi$	0.5 MHz
Cavity dephasing rate	$\kappa_\phi/2\pi$	0.3 MHz
Magnon dephasing rate	$\gamma_\phi/2\pi$	0.3 MHz
Cavity energy lifetime	$T_{1,c} = 1/\kappa_d$	159 ns
Magnon energy lifetime	$T_{1,m} = 1/\gamma_d$	318 ns
Cavity pure dephasing time	$T_{\phi,c} = 1/\kappa_\phi$	530 ns
Magnon pure dephasing time	$T_{\phi,m} = 1/\gamma_\phi$	530 ns
Cavity coherence time	$T_{2,c}$	199 ns
Magnon coherence time	$T_{2,m}$	289 ns

Derived coherence times are computed from the Lindblad rates via T1 = 1/Γ1, Tφ = 1/Γφ, and 1/T2 ' 1/(2T1) + 1/Tφ [17].

III. PHYSICAL GATE REALIZATIONS

Quantum repeater networks rely on elementary gate operations that can be composed to distribute entanglement over long distances. In this section, we explicitly define the gate primitives used throughout the network analysis. The optical mixing of fields is modeled using a general two-mode exchange (beam-splitter) interaction Hamiltonian

$$\mathcal{H}_{\text{int}} = \beta \left(a^\dagger b + a b^\dagger \right) \tag{8}$$

where β is the effective coupling rate and $a^\dagger$ (a) and $b^\dagger$ (b) are the bosonic creation (annihilation) operators of the two modes. In the single-excitation subspace the system dynamics are represented by the unitary evolution operator expressed as

$$\mathcal{U}_{\text{int}}(t) = \begin{pmatrix} \cos(\beta t) & -i\sin(\beta t) \\ -i\sin(\beta t) & \cos(\beta t) \end{pmatrix}. \tag{9}$$

Thus, the same interaction implements either a symmetric mixer (50:50) or a full SWAP depending on the interaction time.

A. BEAM-SPLITTER (50:50)

Following the generation of local entanglement between the respective cavity-magnon modes, the optical field is transmitted through the fiber channel over a distance L (the distance from a node to the midpoint) before being interferometrically combined at a beam splitter. This effectively removes the "which-path" information, leading to maximal mixing of states. In the case of a symmetric beam splitter, we have $|\cos(\beta t)|^2 = |\sin(\beta t)|^2 = \frac{1}{2}$, such that the reflected beam experiences a relative $\pi/2$-phase shift, while the transmitted beam remains unchanged. By using the unitary evolution in (9) and setting the interaction time $t = \pi/(4\beta)$, the input–output relations of the beam splitter are given by

$$|\psi_{\text{out}}\rangle = \mathcal{U}_{\text{int}}\left(t = \frac{\pi}{4\beta} \right) |\psi_{\text{in}}\rangle$$

$$|\psi_{\text{in}}\rangle = \begin{pmatrix} v_1 \\ h_1 \end{pmatrix} \qquad |\psi_{\text{out}}\rangle = \begin{pmatrix} v_2 \\ h_2 \end{pmatrix} \tag{10}$$

where v_1 and h_1 label the two input spatial modes of the beam splitter, and v_2 and h_2 label the corresponding output spatial modes. This 50:50 transformation mixes the two fields as illustrated in Fig. 3(c). The corresponding gate dynamics and fidelity under Lindblad evolution are shown in Fig. 3(a). In the repeater protocol (see Section IV), the beam splitter is followed by photon detection to realize a linear-optical BSM based on two-photon interference. The BSM heralding probability depends on channel transmission and detection efficiencies and is accounted for in the network model (see Section IV).

B. SWAP GATE

The local SWAP operation between the cavity and magnon modes is implemented using the exchange interaction in (4). In the single-excitation subspace, an initial cavity excitation evolves as

$$|\psi(t)\rangle = \cos(g_{mc}t)|10\rangle - i\sin(g_{mc}t)|01\rangle. \tag{11}$$

Choosing the interaction time $t = \pi/(2g_{mc})$ realizes a SWAP-like transfer $|10\rangle \to -i|01\rangle$ (up to a known phase). Dissipation and number-operator dephasing are included via Lindblad collapse operators $\{\sqrt{\kappa_d}\, a, \sqrt{\gamma_d}\, b, \sqrt{\kappa_\phi}\, a^\dagger a, \sqrt{\gamma_\phi}\, b^\dagger b\}$, and the resulting gate dynamics are shown in Fig. 3(b).

Within the repeater protocol, this SWAP gate is used to coherently map quantum states between the cavity mode and the stationary magnon. After a successful heralded entanglement generation step (via the beam-splitter BSM), a cavity–magnon SWAP stores the created entanglement in the magnon mode for subsequent waiting and synchronization across links. At intermediate repeater nodes, the same operation is applied to retrieve the stored excitation back to the cavity mode (or to place it into the cavity as needed) prior to local BSMs for entanglement swapping.

IV. QUANTUM REPEATER NETWORKS

Quantum repeater networks extend the range of quantum communication by mitigating photon loss and decoherence over long channels. In this section, we consider two identical cavity-magnon nodes connected by a fiber link, as illustrated in Fig. 5. Each node is initialized in the same local entangled state $\rho_{\text{loc}} = |\psi^-\rangle\langle\psi^-|$, where $|\psi^-\rangle$ is the Bell state generated according to (5). After local cavity-magnon entanglement generation, the traveling field (optical in the fiber-connected case) propagates a distance L through the channel and is then interfered on a 50:50 beam splitter described by the transformation in (10). A subsequent BSM on the output modes yields an intrinsic (loss-independent) two-outcome Bell discrimination success probability of 50%, corresponding to unambiguous identification of two of the four Bell states. Channel transmission for each photon is modeled using the standard attenuation law $10^{-\alpha L/10}$; these propagation

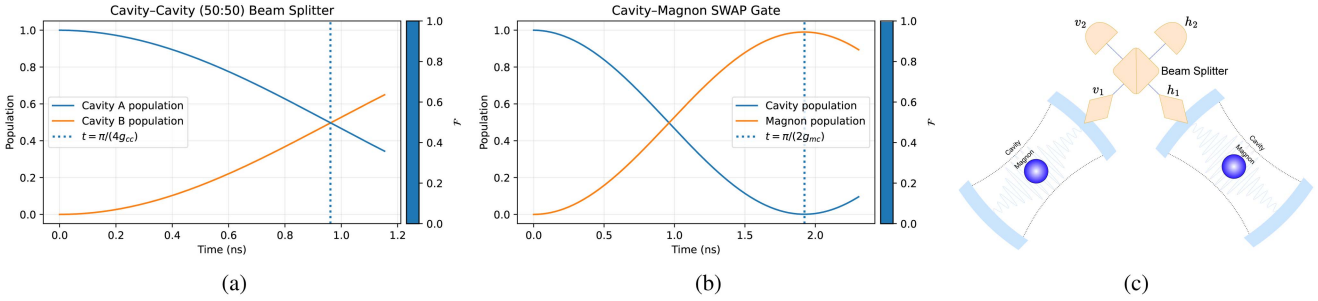


FIGURE 3. Elementary operations used in the repeater protocol, including the effect of noise. (a) Cavity–cavity 50:50 beam-splitter gate, showing populations versus time for an initial $|10\rangle$ state. The dashed line indicates the gate time $t = \pi/(4g_{cc}) = 0.9615$ ns; the right panel reports the corresponding fidelity $\mathcal{F} = 0.993$. (b) Cavity-magnon SWAP gate, showing excitation transfer for an initial $|10\rangle$ state. The dashed line indicates the gate time $t = \pi/(2g_{mc}) = 1.9231$ ns; the right panel reports $\mathcal{F} = 0.990$. (c) Beam-splitter schematic and the associated input–output mode relations used for two-photon interference and the linear-optical BSM in the cavity–magnonic architecture.

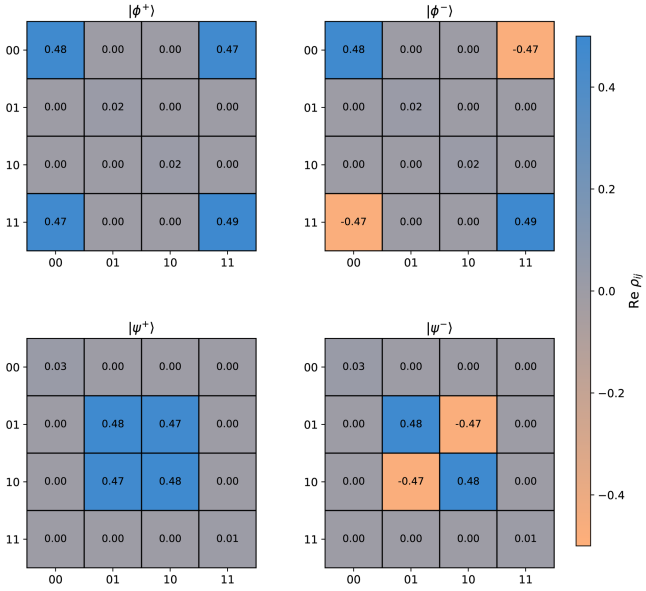


FIGURE 4. Magnon–magnon reconstructed density matrices after entanglement swapping at distant nodes.

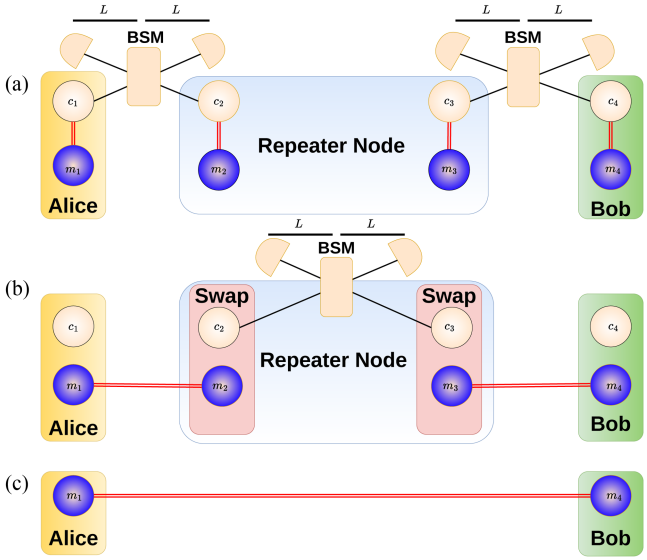


FIGURE 5. Illustration of a cavity–magnon repeater network having cavities c_1, c_2, c_3, and c_4 in pale yellow and magnons m_1, m_2, m_3, and m_4 in blue circles from left to right. Doubled red lines represent entanglement between subsystems, and the beam splitter is positioned at the midpoint of each link, a distance L from each node.

losses, together with readout, conversion (when applicable), and additional parasitic losses, are absorbed into the link efficiency $\eta_{\text{link}}(L)$ defined in Table 3, ensuring consistency between the analytical model and the numerical simulations. Conditioned on a successful click pattern, the measurement provides a heralding signal that projects the remote magnon pair into one of the corresponding entangled outcomes, as depicted in Fig. 4. More generally, Bell-state analyzer performance can exceed the linear-optical 50% limit using boosted or nonlinear Bell state analyzer (BSA) schemes; in our analysis this is captured phenomenologically via an increased $\mathcal{P}_{\text{BSA}}$ (see Table 4) [38].

In Fig. 4, the overall purity of each Bell state exceeds 94%, as the expected basis populations reach about 0.48–0.49, while the matching off-diagonal coherence reaches $\pm$ 0.47, very close to the ideal mark of 0.5. This is due to the noise effect of $\approx$ (0.01–0.03) considered in the system. This confirms that our setup reliably creates and tells apart the four maximally entangled Bell states. The noise turns the perfect Bell pair into a $p = 0.94$ Bell state, which will be further reduced along the length of the fiber optics channel. Furthermore, based on an extension of this methodology, we present a practical and scalable setup for implementing quantum repeaters based on hybrid cavity–magnon systems. The architecture is based on two-step entanglement swapping, i.e., one swap is over the noisy channel and another swap is performed locally at the repeater node as illustrated in Fig. 5. This architecture is designed to facilitate long-distance quantum communication by leveraging the strong coherence properties of magnons and high-quality microwave cavities with localized magnon modes found in ferrimagnetic materials. This enables the robust generation, storage, and swapping of entanglement between distant nodes via fiber links, and subsequently extends it through entanglement swapping operations. We break the complete procedure into steps below.

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TABLE 2. BSM Projectors and Feedforward Pauli Corrections

Outcome	$\rho^j = \lvert\Phi_j\rangle\langle\Phi_j\rvert$	$\mathcal{U}_{\text{cor}}^{(i)}$
$\lvert\psi^+\rangle = \frac{1}{\sqrt{2}}(\lvert 01\rangle + \lvert 10\rangle)$	ρ^0	σ_z
$\lvert\psi^-\rangle = \frac{1}{\sqrt{2}}(\lvert 01\rangle - \lvert 10\rangle)$	ρ^1	$\mathbb{I}$
$\lvert\phi^+\rangle = \frac{1}{\sqrt{2}}(\lvert 00\rangle + \lvert 11\rangle)$	ρ^2	$\sigma_z\sigma_x$
$\lvert\phi^-\rangle = \frac{1}{\sqrt{2}}(\lvert 00\rangle - \lvert 11\rangle)$	ρ^3	σ_x

TABLE 3. Analytical Expressions Used in the Repeater-Chain Simulations

Symbol	Meaning	Formula
$\eta_{\text{link}}(L)$	Single-photon link efficiency	$10^{-\alpha L/10}\,\eta_{\text{read}}\,\eta_{\text{conv}}\,\eta_{\text{extra}}$
$\mathcal{P}_{\text{click}}$	Raw Bell-click probability (one channel)	$\mathcal{P}_{\text{BSA}}\left(\eta_{\text{link}}(L)\,\eta_{\text{col}}\,\eta_{\text{det}}\right)^2$
$\mathcal{P}_{\text{succ}}^{(\text{hop})}$	Per-hop success with multiplexing	$1-\left(1-\mathcal{P}_{\text{click}}\right)^{M_{\text{mux}}}$
$\mathcal{P}_h$	Cumulative success after h hops	$\prod_{j=1}^{h}\mathcal{P}_{\text{succ}}^{(j)}$
$\mathcal{F}$	Conditional magnon–magnon fidelity	$\langle\psi^-\rvert\rho_{m_1 m_{h+1}}\lvert\psi^-\rangle$

A. PREPARATION

We consider four different nodes, and each node is initialized in an identical singlet Bell state as discussed above. The mathematical expression for the density matrix is

$$\rho_{\text{loc}} = \bigotimes_{n=1}^{4} \rho_{c_n m_n}$$

where $\rho_{c_n m_n} = p|\psi^-\rangle\langle\psi^-|_n + (1-p)\rho_{\text{noise}}$ is the density matrix of the singlet Bell state generated between the cavity–magnon of each nth node. Here, p is the purity of the singlet Bell state, which is reduced due to the noise accumulating with length and each new interaction, thus leading to a mixed state with $1-p$ probability.

B. FIRST-STAGE ENTANGLEMENT SWAPPING

Photons from cavities c_1 and c_2 and similarly from c_3 and c_4 interfere pairwise on a beam splitter and are detected to perform a BSM, as shown in Fig. 5(a). Each photonic BSM heralds the corresponding magnon pairs, (m_1, m_2) and (m_3, m_4) onto one of the four Bell states. In practice, the particular Bell state obtained is random and determined by the BSM outcome. At this point, a classical feedforward channel conveys the two-bit Bell outcome to the repeater node and the end node, and a conditional Pauli correction $\mathcal{U}_{\text{cor}}^{(i)} = \sigma_i$, where $i \in \{0, x, y, z\}$ and $\sigma_0 = \mathbb{I}$, is applied so that in all cases the remaining two-magnon state rotates into the singlet as shown in Table 2. For simplicity, we consider the case where both measurements project their magnon pairs onto the singlet state $|\psi^-\rangle\langle\psi^-|$. Under this assumption, no correction is needed, and the joint magnonic state at the repeater node becomes $\rho = \rho_{m_1,m_2} \otimes \rho_{m_3,m_4}$ with $\rho_{m_i m_j} = |\psi^-\rangle\langle\psi^-|_{m_i,m_j}$, giving us adjacent magnon–magnon entanglement.

C. FINAL SWAP GATE AT REPEATER NODE

At the repeater node, the local cavities c_2 and c_3 are initialized in their vacuum states $|0\rangle_{c_2}|0\rangle_{c_3}$. The quantum information stored in magnons m_2 and m_3 is then swapped into their respective cavities by the SWAP gate using the exchange interaction in (4) for an interaction time $t = \pi/(2g_{mc})$.

Furthermore, a beam-splitter operation is performed between cavities c_2 and c_3. A subsequent BSM on these two cavities then heralds the successful generation of end-to-end entanglement between the distant magnons m_1 and m_4 after applying the unitary correction as shown in Table 2 for feedforward as discussed in [39]. This cancels any mismatch, mitigates any additional losses, and adds to the total success probability of the system. Another factor that adds to the probability of success is the multiplexing M of the entangled state in the fiber channel. We consider different levels of multiplexing in the fiber channel based on the experimentally relevant literature [40], [41], [42], [43]. Finally, a reset signal brings all the cavities back to the initialized state and repeats the process for n number of nodes.

V. RESULTS AND DISCUSSION

To evaluate the practical feasibility of our cavity–magnon quantum repeater architecture, we perform extensive numerical simulations that account for realistic experimental conditions and system imperfections using the analytical expressions provided in Table 3 and the parameter sets detailed in Table 4. Our analysis evaluates end-to-end performance across multiple network hops, considering both the quality of distributed entanglement and the probability of successful operations. We quantify repeater performance using three key metrics, concurrence as a measure of entanglement strength between distant magnons, fidelity comparing the final distributed state to the ideal Bell state, and success probability accounting for all sources of loss and measurement failures throughout the protocol.

Our simulations model a 20-hop quantum repeater chain, where each hop represents the elementary link between adjacent network nodes. We consider two distinct deployment scenarios: chip-scale implementation with $L = 1$ cm fiber links operating entirely in the microwave domain, and metro-scale deployment with $L = 10$ km fiber links requiring microwave-to-optical conversion. The total success probability incorporates multiple loss mechanisms. For a single photon traversing a span of length L, the link efficiency is $\eta_{\text{link}}(L) = 10^{-\alpha L/10}\,\eta_{\text{read}}\,\eta_{\text{conv}}\,\eta_{\text{extra}}$, where α is the channel attenuation, η_{read} is the local readout efficiency, η_{conv} is the microwave-to-optical conversion efficiency (when applicable), and η_{extra} captures additional parasitic losses. The raw Bell-click probability for a single spectral channel is then $\mathcal{P}_{\text{click}} = \mathcal{P}_{\text{BSA}}\left(\eta_{\text{link}}(L)\,\eta_{\text{col}}\,\eta_{\text{det}}\right)^2$, explicitly accounting for Bell-state analyzer efficiency $\mathcal{P}_{\text{BSA}}$, collection efficiency η_{col}, detector efficiency η_{det}, and link loss.

Fig. 6(a) presents the performance characteristics for chip-scale deployment with 1 cm links. Starting from an initial

TABLE 4. Parameter Set Used in the Cavity–Magnon Repeater Chains

Scenario	α	L_{span}	η_{read}	η_{conv}	$\text{loss}_{\text{extra}}$	η_{det}	η_{col}	$\mathcal{P}_{\text{BSA}}$	M_{mux}
Chip-A (1 cm)	$0.20\,\text{dB cm}^{-1}$	1 cm	0.62	–	0.98	0.98	0.95	0.50	1
Chip-B (1 cm, multiplexed)	$0.20\,\text{dB cm}^{-1}$	1 cm	0.62	–	0.98	0.98	0.95	0.50	8
Chip-C (1 cm, ancilla, dense multiplexed)	$0.20\,\text{dB cm}^{-1}$	1 cm	0.62	–	0.98	0.98	0.95	0.75	30
Metro-A (10 km)	$0.35\,\text{dB km}^{-1}$	10 km	0.62	0.005	0.90	0.80	0.95	0.50	1
Metro-B (10 km, multiplexed)	$0.20\,\text{dB km}^{-1}$	10 km	0.62	0.50	0.95	0.98	0.95	0.50	8
Metro-C (10 km, dense multiplexed)	$0.16\,\text{dB km}^{-1}$	10 km	0.62	0.80	0.95	0.98	0.95	0.75	30

Chip-Scale attenuation α is given in dBcm−1 and span length L_{span} in centimeters; metro-fiber values are in dBkm−1 and kilometers, respectively. Here, L_{span} denotes the node-to-midpoint distance. A dash "–" indicates that microwave–to–optical conversion is not present (purely microwave link) [44]

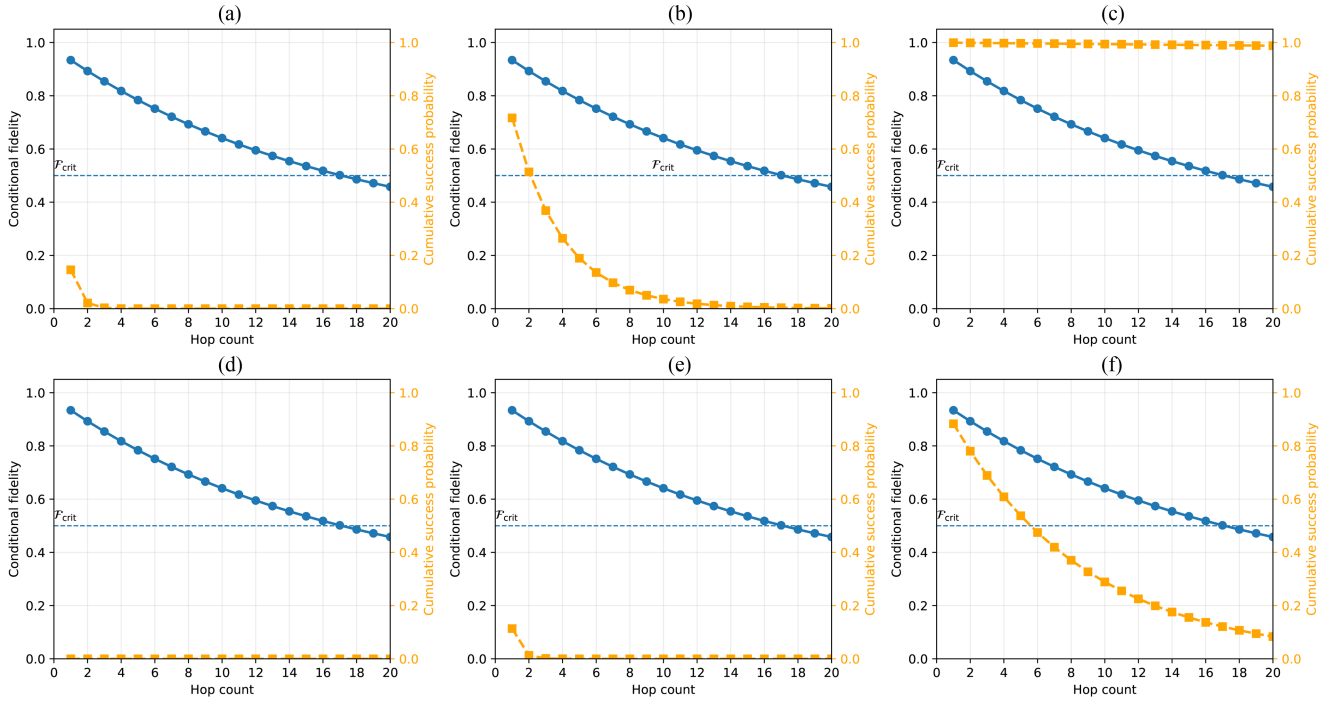


FIGURE 6. Performance comparison of entanglement distribution for on-chip (1 cm) and metro-scale (10 km) links under different multiplexing configurations. (a)–(c) On-chip cases: (a) Chip-A (single channel), (b) Chip-B (multiplexed), and (c) Chip-C (dense multiplexed). (d)–(f) Metro-scale cases: (d) Metro-A (single channel), (e) Metro-B (multiplexed), and (f) Metro-C (dense multiplexed).

Bell state with 0.97 concurrence, our simulations demonstrate a gradual but systematic degradation of entanglement quality with hop count; the conditional fidelity decreases from $\mathcal{F} \approx 0.93$ after the first swapping operation to $\mathcal{F} \approx 0.46$ by 20 hops. Relative to the critical fidelity threshold $\mathcal{F}_{\text{crit}} = 0.5$ indicated in Fig. 6, the chain remains above threshold until approximately 16 hops and then falls below. However, the success probability follows a more dramatic progression, with Chip-A exhibiting a rapidly vanishing cumulative success probability after only a few hops (dropping from $\mathcal{P}_h \approx 0.15$ at the first hop to essentially zero by ∼3–4 hops). This steep decline primarily results from the cumulative effect of BSM inefficiencies, where each 50% success probability compounds across multiple operations.

The introduction of eight-channel multiplexing improves performance characteristics, as demonstrated in Fig. 6(b). The per-hop success probability increases by exploiting parallel operation across multiple spectral channels. The multiplexed success probability follows $\mathcal{P}_{\text{succ}}^{(\text{hop})} = 1 - \left(1 - \mathcal{P}_{\text{click}}\right)^{M_{\text{mux}}}$, where M_{mux} represents the number of multiplexed channels. For Chip-B, the cumulative success probability is substantially enhanced at short range (starting at $\mathcal{P}_h \approx 0.7$ at hop 1), but still decays rapidly with hop count, reaching $\mathcal{P}_h \approx 10^{-2}$ by about 10 hops and becoming effectively negligible by 20 hops. Notably, the conditional fidelity trend is only weakly affected by multiplexing, following the same hop-dependent decay and reaching $\mathcal{F} \approx 0.46$ by 20 hops. Fig. 6(c) illustrates the performance under dense multiplexing conditions with 30 channels and advanced Bell-state analyzers achieving 75% efficiency. In Chip-C, the cumulative success probability remains near unity across all 20 hops ($\mathcal{P}_h \approx 1$ over the full range), while the conditional fidelity still decreases with hop count, crossing $\mathcal{F}_{\text{crit}}$ at approximately the same hop number as Chip-A/B. This regime represents the theoretical upper bound for the heralding rate

in our architecture, demonstrating the scalability potential of success probability when combined with dense multiplexing and improved BSA efficiency.

Metro-scale deployment introduces additional complexity through the requirement for microwave-to-optical conversion. Fig. 6(d) shows the performance with realistic conversion efficiency $\eta_{\text{conv}} = 0.005$, representative of current experimental capabilities. In this case, the cumulative success probability is effectively zero across the plotted hop range, indicating that present-day conversion efficiencies are a dominant bottleneck for multihop metro-scale operation. However, Fig. 6(e) and (f) demonstrates the impact of improved conversion technologies. With moderate efficiency improvements ($\eta_{\text{conv}} = 0.5$; Metro-B), the first-hop success rises but still collapses rapidly with hop count, leaving $\mathcal{P}_h \approx 0$ beyond the first few hops. The highest performance scenario ($\eta_{\text{conv}} = 0.8$; Metro-C) maintains a nonnegligible cumulative success probability over many hops, decreasing from $\mathcal{P}_h \approx 0.9$ at hop 1 to $\mathcal{P}_h \approx 0.1$ at hop 20, suggesting that advances in microwave-to-optical interfaces will be crucial for practical metro-scale deployments.

The $L = 10$ km fiber links introduce additional photon loss through standard telecom fiber attenuation ($\alpha = 0.16$–0.35 dB/km). Our simulations show that this distance-dependent loss, while significant, can be compensated through multiplexing and improved detector efficiency, particularly when combined with high-efficiency conversion technologies. The relationship between multiplexing level and system performance follows predictable scaling laws, where for M_{mux} multiplexed channels, the effective success probability scales as $1 - (1 - \mathcal{P}_{\text{click}})^{M_{\text{mux}}}$, with $\mathcal{P}_{\text{click}}$ representing the single-channel success probability. Consistent with Fig. 6, the gain from moving from single-channel operation to eight-channel multiplexing is substantial at short hop counts (Chip-A $\rightarrow$ Chip-B), while dense multiplexing (30 channels) can effectively saturate the heralding probability (Chip-C) and maintain nonnegligible success rates even for metro-scale links (Metro-C).

Dense multiplexing imposes practical constraints on system complexity, requiring sophisticated wavelength division multiplexing infrastructure and high-bandwidth detection systems for the 30-channel implementation [45], [41], [42]. However, our analysis suggests that moderate multiplexing (eight channels) captures most of the performance benefits while maintaining reasonable implementation complexity. While our analysis focuses on spectral multiplexing through wavelength division, the framework extends naturally to temporal multiplexing schemes, with recent experiments demonstrating both approaches where spectral multiplexing offers advantages for continuous operation and temporal multiplexing provides simpler hardware requirements.

Quantum communication protocols typically require high conditional fidelity (often $\mathcal{F} \gtrsim 0.7$, depending on the application). In our 20-hop simulations, this regime is maintained only up to a finite hop count: the conditional fidelity drops to $\mathcal{F} \approx 0.7$ at roughly $\sim$8 hops and then decreases monotonically, crossing the critical threshold $\mathcal{F}_{\text{crit}} = 0.5$ near $\sim$16 hops (see Fig. 6). This margin provides robustness against additional sources of error not captured in our idealized simulations for short-to-intermediate hop counts. From a systems perspective, cavity–magnon nodes offer strong integration advantages relative to atomic-ensemble and NV-center repeater platforms: they operate naturally in the microwave domain (matching superconducting processors), provide wide frequency tunability, and can interface with fiber networks via microwave-optical transduction [46], [47]. A full quantitative cross-platform benchmark is beyond the scope of this work; instead, we provide a hardware-consistent baseline that quantifies the tradeoff between conditional fidelity, heralding probability, and hop count under experimentally motivated noise parameters. The hybrid nature of our platform enables interfaces with both superconducting and optical quantum systems.

Beyond optical loss and BSM heralding, Fig. 6 also highlights an operational bottleneck associated with the finite entanglement-storage window: as hop count increases, the number of sequential swapping attempts required to obtain a simultaneous set of heralds can grow, effectively tightening the allowable memory lifetime for the end nodes. In practice, this motivates operating in a regime where heralding is made near-deterministic (Chip-C and Metro-C) and where classical feedforward and reinitialization latencies are minimized. These requirements can be addressed without changing the core protocol by 1) increasing the effective per-hop success probability via spectral/temporal multiplexing and improved BSA efficiency, 2) reducing per-attempt cycle time through faster local gates and integrated control electronics, and 3) employing standard nested scheduling and buffering strategies so that successfully prepared elementary links can be stored while awaiting the remaining heralds. For metro-scale links, the time for the photon to reach the midpoint for BSM and for the classical herald to return exceeds the magnon coherence time of 289 ns in Table 1. Current YIG coherence times are therefore insufficient for the heralding latency of a 10 km link, and the Metro-B and Metro-C results should be read as contingent on future improvements in magnon coherence, or on transferring the heralded excitation into a longer-lived buffer mode. The chip-scale scenarios are unaffected, since at this scale the propagation latency is negligible relative to the coherence budget.

The heralded success probabilities translate directly to communication throughput, where Fig. 6 highlights a clear separation between fidelity-limited and success-limited operating regimes: even when fidelity remains acceptable over many hops, the cumulative success probability can become the limiting factor (Chip-A/B, Metro-A/B). Conversely, Chip-C (on-chip) and Metro-C (metro-scale with improved conversion) sustain substantially higher success probabilities over longer chains, enabling higher effective entanglement distribution rates at fixed clock cycles. Our parameter sets derive directly from state-of-the-art experimental demonstrations, ensuring that the simulated performance

reflects achievable hardware capabilities. The YIG sphere parameters (Q-factors $10^4 - 10^6$, coupling rates 130 MHz) represent current experimental standards, while the projected improvements in conversion efficiency align with ongoing research trajectories [46], [47].

The modular architecture of our repeater design enables straightforward network expansion, where additional hops follow the same scaling laws with performance degrading predictably according to the cumulative loss model. This predictability facilitates network planning and resource allocation for large-scale deployments. Real implementations must address several practical challenges not captured in our idealized simulations, including frequency stabilization across network nodes, synchronization of entanglement swapping operations, and integration with classical control systems for feed-forward operations. However, the magnonic platform's inherent stability and controllability provide advantages for addressing these challenges.

Our analysis employs several simplifying assumptions that allow tractable modeling of the system. The Lindblad master equation approach provides a phenomenological treatment of decoherence but may not capture all environmental effects, and the assumption of perfect spatial mode matching between cavity and magnon modes represents an idealization that requires careful experimental optimization. In addition, our metro-scale simulations assume the heralded entanglement can be stored for the full link latency, an assumption not met by current magnon coherence times. Several technical hurdles must be overcome for practical implementation, particularly the development of high-efficiency, low-noise microwave-to-optical converters, which remains a critical bottleneck for metro-scale deployment. In addition, the realization of high-fidelity Bell-state analyzers with deterministic outcomes would dramatically improve system performance.

VI. CONCLUSION

In this article, we developed and analyzed a comprehensive cavity–magnon quantum repeater architecture for scalable quantum communication networks. Through extensive numerical simulations incorporating realistic experimental parameters, we quantified end-to-end performance over multihop chains and identified the operating regimes where cavity–magnon nodes can support scalable entanglement distribution for both chip-scale and metro-scale deployments. Our cavity–magnon platform offers unique advantages, including excellent frequency tunability, reduced environmental sensitivity, and seamless integration with both superconducting and optical quantum systems, positioning it as an ideal interface for heterogeneous quantum networks.

Our analysis reveals that while chip-scale implementations achieve high-fidelity operation over a finite hop budget, the overall throughput is ultimately limited by heralding statistics unless multiplexing is employed. In particular, dense multiplexing can drive the per-hop success toward near-deterministic operation, whereas moderate multiplexing provides substantial gains with lower hardware overhead. Metro-scale deployment faces critical bottlenecks in microwave-to-optical conversion efficiency. The development of high-efficiency, low-noise converters is a primary technological challenge that must be addressed for practical long-distance quantum networks. In addition, the finite entanglement-storage window imposes timing constraints as hop count increases, motivating architectures that minimize cycle time and reduce the number of sequential attempts required to obtain simultaneous heralds; in particular, the metro-scale results are contingent on magnon coherence times long enough to cover the heralding latency of long fiber links. In addition, practical implementation requires advances in frequency stabilization across distributed nodes, synchronization of entanglement swapping operations, and integration with classical control systems for real-time network management.

The scalability of our architecture, through spectral multiplexing, provides a clear pathway for performance enhancement while maintaining reasonable implementation complexity. Building on the trends observed in our multihop simulations, future work should prioritize 1) converter technology development that simultaneously improves efficiency and suppresses added noise, 2) system-level co-design of multiplexing, detection, and feed-forward latency to satisfy memory-lifetime constraints, and 3) protocol scheduling and buffering strategies that better utilize successful elementary links. Future research should prioritize conversion technology development through electro-optic and optomechanical approaches, exploration of hybrid multiplexing schemes, and investigation of alternative network topologies beyond linear chains. Integration with quantum error correction protocols represents another crucial direction for enabling fault-tolerant distributed quantum computation.

Our cavity–magnon quantum repeater architecture establishes a technologically well-motivated foundation for next-generation quantum networks. Its compatibility with existing fiber-optic infrastructure and its clear scaling behavior under multiplexing and improved conversion efficiencies position these systems as a viable bridge between laboratory demonstrations and practical quantum communication applications. As quantum information technologies transition toward real-world deployment, our work provides both the theoretical framework and an experimentally informed roadmap for implementing scalable quantum repeater networks that can support the emerging quantum internet infrastructure.

DECLARATIONS

ETHICAL APPROVAL

We, the undersigned, hereby declare that this manuscript contains original work and has not been published before, nor is it currently being considered for publication elsewhere.

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Appropriate references have been made for any information drawn from other work.

COMPETING INTERESTS

We confirm that there are no known conflicts of interest associated with this publication, and that no author has any conflict that could affect the results or the timing of publication.

AUTHORS' CONTRIBUTIONS

The manuscript has been read and approved by all named authors, and there are no other persons who satisfy the criteria for authorship but are not listed. We confirm that all authors have approved the order in which they are listed in the manuscript. We confirm that we have given due consideration to the protection of intellectual property associated with this work and that there are no obstructions to publication.

AVAILABILITY OF DATA AND MATERIALS

The data supporting the findings of this study are available from the corresponding author upon reasonable request.

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